

SYSTEM ARCHITECTURE DOCUMENT (SAD) TEMPLATE

Reference Template for Document Validation System

1. Introduction

1.1 Purpose

State the explicit purpose of this System Architecture Document (SAD). Provide a comprehensive technical blueprint of the software platform's structural components, design patterns, and deployment patterns.

1.2 System Scope

Define the technical boundaries of the document, declaring exactly which cloud infrastructure networks, application domains, microservices, and external systems are addressed.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SAD	System Architecture Document
API	Application Programming Interface
IAM	Identity and Access Management

1.4 References

List foundational requirements source documents including the Functional Requirements Document (FRD) and explicit Non-Functional Requirements (NFR) matrices.

1.5 Document Overview

Outline how the architectural views and schemas are organized across this document.

2. Conceptual High-Level Design

2.1 Architecture Diagram View

Provide or reference a visual block diagram mapping core sub-systems, frontend presentation layers, container orchestration boundaries, and external API dependencies.

2.2 System Component Registry

Component ID	Subsystem Name	Functional Responsibility
ARCH-01	UI Frontend Layer	Serves client web components and captures user interaction payloads.
ARCH-02	Core API Gateway	Validates security headers, handles rate-limiting, and controls routing.

3. Core Tech Stack and Integration Blueprints

3.1 Software and Hardware Interface Specifications

Explicitly declare core database systems (SQL/NoSQL structures), hosting platforms (AWS, GCP, or Azure environments), and third-party software service integration channels.

3.2 Data Topography and Storage

Detail explicit data persistency schemas, caching distribution tiers (e.g., Redis), object storage policies, and synchronous vs. asynchronous data flow mechanisms.

4. Non-Functional Architectural Provisions

4.1 Security Blueprint

Detail transport encryption protocols (TLS 1.3), encryption-at-rest keys, Identity and Access Management (IAM) role limitations, and network perimeter firewall rules.

4.2 Scalability and Resiliency Rules

Outline load balancing configurations, multi-region database replication schedules, automatic horizontal scaling rules, container failover protocols, and target performance recovery indices.

5. Validation Rules

5.1 Required Sections

- Introduction
- Conceptual High-Level Design
- Core Tech Stack and Integration Blueprints
- Non-Functional Architectural Provisions
- Validation Rules
- Appendices

5.2 Validation Criteria

The compliance engine must verify the presence of explicit component registries, structural network layouts, and comprehensive security encryption strategy breakdowns.

6. Appendices

6.1 Appendix A

Detailed entity-relationship diagrams (ERDs), infrastructure-as-code configuration scripts, or network topology scheme diagrams.